

EXAM 7 – SPRING 2014

9. (2 points)

Given the following output (\$000) from a paid chain-ladder generalized linear model:

Accident <u>Year</u>	Mean <u>Unpaid</u>	Standard <u>Error</u>	Coefficient <u>Of Variation</u>
2008	\$63	\$35	0.556
2009	632	260	0.411
2010	2,130	641	0.301
2011	4,721	1,265	0.268
<u>2012</u>	<u>7,011</u>	<u>2,825</u>	<u>0.403</u>
Total	14,557	4,024	0.276

Use four diagnostic tests to evaluate the model for reasonableness.

CONTINUED ON NEXT PAGE

SAMPLE ANSWERS AND EXAMINER'S REPORT

QUESTION: 9	
TOTAL POINT VALUE: 2	LEARNING OBJECTIVE: A9: Test assumptions underlying reserve models.
SAMPLE ANSWERS	
<i>Sample Answer 1</i> Standard Error should increase over time, oldest to youngest; it does. This is due to the larger reserve values at more recent years Standard error is increasing over time – this is reasonable <i>Sample Answer 2</i> Coefficient of variation decreases as AY increases, except for the most recent AY. This is good. It decreases since the variability is measured against a higher mean. The last AY has higher coefficient of variation because the parameter uncertainty overpowers process uncertainty. Does CV decrease as AY becomes less mature? In general yes. Also it is reasonable to see an uptick for the most recent year because sometimes the parameter variance increases enough to overpower the generally declining CV ratio <i>Sample Answer 3</i> CV should decrease in more recent AYs due to influence of incremental claim payments. This test is not passed as CV for AY 2012 is greater than CV for AY's 2010 and 2011 CV should be lower for more recent AY's – but $.403 > .268$, so the model is not reasonable in this sense <i>Sample Answer 4</i> Standard error of the total reserve should be larger than any individual year, and it is The total standard error should be greater than the standard error for any one period <i>Sample Answer 5</i> CV of the total reserve should be less than any individual year, and it is not (it is larger than 2011). This should be investigated further CV total should be lower compared to CV in any individual AY. 2011 violates this <i>Sample Answer 6</i> The mean reserve should show an increasing pattern down the column, which it does Mean unpaid → should increase as we move down the AY's. This occurs, so passes.	

SAMPLE ANSWERS AND EXAMINER'S REPORT

EXAMINER'S REPORT

The candidate was expected to know the diagnostic tests, then apply them to the data.

Relatively few candidates received full credit. Common errors fell into the following four buckets

- Diagnostic tests that weren't accepted as valid response, some common ones:
 - The total standard error should be less than the sum of the individual standard errors: this is expected to be true of the output of the model, but it will also be true of the output of an unreasonable model.
 - Similar responses to prior bullet, but discussing coefficient of variation or sum of squared errors, were also not accepted
 - There were several variations of the following response: "compare the standard deviation to the mean, in general we want the mean to be at least two times the standard deviation." However, a symmetric normal distribution may not be a good approximation to the true distribution of the reserves. This diagnostic would only assess whether the distribution of reserves in the model can be reasonably described as normal, not assess the reasonableness of the model itself.
- No assessment of a correctly described diagnostic test
- Incorrect assessment of a correctly described diagnostic test: one surprising example was related to accepted response 5. The correct diagnostic was described, but failure of AY 2011 not noted
- Diagnostic tests not appropriately described: most common for Accepted Response3: "CV should be high in early accident years and latest year" this response did not describe whether the expectation is to have a decreasing or increasing CV

This question was challenging from the standpoint of synthesizing the tests and applying it to the data, in addition to clearly stating what the diagnostic tests were.